GGG-V-445a
August 3, 1966

SUPERSEDING
Int. Fed. Spec. GGG-V-00445 (GSA-FSS)
March 13, 1965 and
Fed. Spec. GGG-V-436d
June 20, 1958 (In part) (See 6.2)

FEDERAL SPECIFICATION

VISE, WOODWORKER'S AND PATTERNMAKER'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vises for holding various sizes and shapes of wood and are suitable for bench or table mounting.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Vises covered by this specification shall be of the following types and classes as specified (see 6.1):

Type I—Woodworker's.

Class 1—Continuous screw.
Class 2—Quick acting.

Type II—Patternmaker's.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposals, form a part of this specification.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

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3.1 **Illustrations.** The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of vises otherwise conforming to this specification.

3.2 **Materials.** Materials used in the construction of vises shall be as hereinafter specified. All castings shall be of uniform quality, free of blowholes, porosity, hard spots, shrinkage defects, cracks, or other injurious defects. Strength and other essential physical properties of the casting shall be adequate throughout for the purpose intended. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended.

3.3 **Coatings.** All unmachined surfaces of the component parts of vises shall be given either one coat of baked enamel or lacquer. The color may be at the manufacturer's option. In no case shall fillers be used.

3.4 **Identification marking.** Each vise shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.5 **Interchangeability of parts.** All parts having the same manufacturer's number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.6 **Vise screw.** The vise screw shall be made from steel of not less than 50,000 pounds per square inch (p.s.i.) yield, and 65,000 p.s.i. tensile strength (see 4.4.1).

3.7 **Type I, woodworker's vise.**

3.7.1 *Class 1, continuous screw.* Class 1 vise shall consist of a body with back jaw, front jaw, screw, nut, guide rods, screwhead, drag bar, and handle. The vise shall have a continuous-action screw and nut. The vise shall withstand the test specified in 4.4.

3.7.1.1 *Body.* The body shall be of cast iron of solid construction, consisting of a back jaw, bench mounting, and rear bearing sections. The back jaw section shall have at least two drilled holes in the upper portion of the jaw face for mounting of a wood facing. The lower portion of the back jaw shall have the front guide-rod bearings and vise nut mounting. The bench mounting section shall be located on the rear side of the back jaw section and shall be square with the jaw face within 2° in either direction. At least two bench mounting holes shall be provided. The rear bearing section shall be so located and have bearing holes of such fit as to permit opening the jaws to their extreme capacity without bind.

3.7.1.2 *Front jaw.* The front jaw shall be of cast iron and shall be permitted to incline a maximum of 1/16 inch from vertical toward the top of the back of the jaw to compensate for any spring of the parts in clamping of work. The two guide rods shall be rigidly anchored to the bottom portion of the front jaw and the screw bearing shall be fastened between the guide rods. Four holes shall be provided for the mounting of a wood jaw facing. The front jaw shall be provided with at least one 1/2- by 1-inch brass or soft steel dog or stop.

3.7.1.3 *Screw.* The screw shall be in accordance with 3.6 with a minimum diameter of 1 inch. It shall have a shoulder or collar and bearing to fit the front jaw, and shall terminate in the screwhead on the front end and in a bearing in the drag bar on the other. The bearing surfaces on the screw shall be of such fit and the screwhead so located with reference to the shoulder or collar as to permit the screw to rotate freely. The continuous-thread-type screw may be either single or double threaded, and shall be so designed that two revolutions of the double-thread screw or four complete revolutions of the single-thread screw shall move the front jaw a distance of 1 inch.

3.7.1.4 *Nut.* The nut shall be of malleable

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iron, cast iron, ductile iron, steel, or bronze threaded to engage the screw, and shall be renewable.

3.7.1.5 *Guide rods.* The rods shall be parallel to the axis of the screw, shall be securely fastened to the front jaw and extended through the holes in the body. The minimum diameter of guide rods shall be 7/8 inch.

3.7.1.6 *Screwhead.* The screwhead shall be of cast iron or malleable iron and shall be securely fastened to the screw. The screwhead shall be provided with a hole at right angles to the axis of the screw to receive the handle. The hole for the handle shall be chamfered or otherwise shaped at both ends to prevent accidental pinching of the hands. The screwhead for adjustable steel handles shall be provided with a tension spring to hold the handle at any desired location.

3.7.1.7 *Drag bar.* The drag bar shall be of cast iron, malleable iron, or steel with holes drilled at both ends to fit the rear end of the guide rods. These holes shall be in alignment with those specified for the front jaw and body through which the guide rods are inserted. The hole in the center of the drag bar shall act as the rear bearing for the vise screw. The drag bar shall be fastened to the vise screw or to the guide rods.

3.7.1.8 *Handle.* The handle shall be of straight-grained hickory, solid aluminum, or steel. It shall be capped to prevent sliding out of the screwhead, or steel bar handles may have machine-type handle grips, securely mounted at right angles near each end. Overall length of wood handle shall be not less than 9 inches, and for metal handle, not less than 7 inches. The handle shall be of adequate strength and diameter to withstand the intended duty.

3.7.1.9 Class 1 vise shall conform to table I for the size specified (see 6.1), and shall be similar to figure 1.

TABLE I. *Type I, woodworker's, classes 1 and 2, continuous screw and quick-acting.*

Jaw size (nominal)	Jaw dimensions (nominal)		Jaw opening (min.)	Weight (min.)
	Width	Depth		
Inches	*Inches*	*Inches*	*Inches*	*Pounds*
4 by 7	7	4	9	25
4 by 10	10	4	12	30

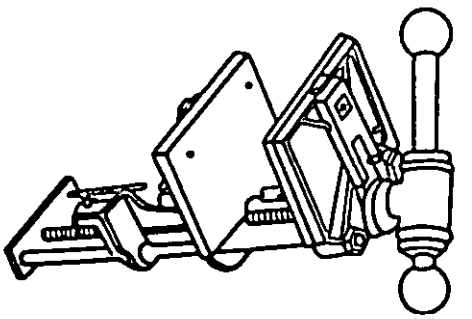


FIGURE 1. Type I, woodworker's, classes 1 and 2, continuous screw and quick-acting.

3.7.2 *Class 2, quick acting.* Class 2 vise shall be in accordance with class 1 vise (see 3.7.1), except it shall be provided with a device that will permit rapid adjustment of the jaw opening with the final pressure or release facilitated by the rotation of the screw in the conventional manner.

3.8 Type II, patternmaker's vise. Type II vise shall consist of a body, square guide beam, front jaw, back jaw, tilt jaw, tilt support with vise lock, screw, and screw handle. The vise shall withstand the test specified in 4.4.2.

3.8.1 *Body.* The body shall be of cast iron and consist of cast housing and locking lever. Housing shall be bored to permit working fit of back jaw boss, allowing full 360° rotation of jaws. Locking lever shall lock jaws at any angle of rotation.

3.8.2 *Square guide beam.* The square guide beam shall be of cast iron and serve as a housing for the screw, and have a free sliding fit through the back jaw for the

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complete opening of the vise without binding. The square guide beam shall extend through both jaws and be securely fastened to the front jaw.

3.8.3 *Front jaw.* The front jaw shall be of cast iron, rigid, and parallel to the back jaw. The front jaw shall be capable of tilting 5° in either direction about its vertical center line (or axle) — with a movement of a cam lever. Two holes shall be provided for the mounting of a soft facing jaw. Either the front or back jaw shall have a horizontal machined groove for placement of the tilt jaw

3.8.4 *Back jaw.* The back jaw shall be of cast iron, rigid, and parallel to the front jaw. It shall have a large boss with machined opening permitting sliding fit of the beam. The outer surface of the boss shall be machined to permit 360° rotation of the back jaw beam and the front jaw in the body.

3.8.5 *Tilt jaw.* The tilt jaw shall be of cast iron and shall be attached, when needed, to the front or back jaw for gripping of angular work. The jaw shall tilt 15° up or down.

3.8.6 *Tilt support for vise with lock.* The tilt support shall be of cast iron, and at least three holes shall be provided for mounting the vise to a bench top. It shall support the vise body on pivot pins to allow full 90° tilt of the vise from the normal horizontal position to a vertical position. A locking device shall permit the vise to be locked at any angle of tilt.

3.8.7 *Screw.* The vise screw shall have American National, square, acme, modified acme, or buttress threads and be in accordance with 3.6. One end of the screw shall have a steel head of adequate size which shall be either an integral part of the screw or welded to the screw in a permanent manner. The head shall be provided with a hole to receive the screw handle. The hole for the screw handle shall be chamfered, or otherwise shaped at both ends, to prevent accidental pinching of the hands. The screw shall engage the full length of the threaded bearing of the body when the jaw is at the maximum opening indicated in the applicable table.

3.8.8 *Screw handle.* The screw handle shall be of hardwood and capped at each end to prevent sliding out of the screwhead. Handle shall be not less than 7 inches in overall length, and shall be of adequate strength and diameter to withstand the duty to be performed.

3.8.9 Type II vise shall conform to table II for the size specified (see 6.1), and shall be similar to figure 2.

TABLE II. *Type II, patternmaker's.*

Jaw size (nominal)	Jaw dimensions		Jaw opening (minimum)	Weight (minimum)
	Depth ±1/2	Width ±1-1/2		
Inches	*Inches*	*Inches*	*Inches*	*Pounds*
5 by 14	5	14	12	55
7 by 18	7	18	14	85

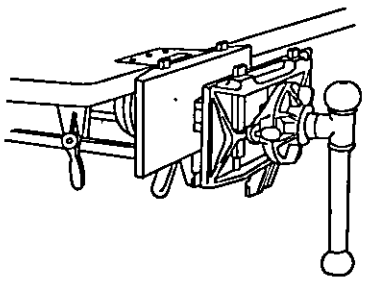


FIGURE 2. Type II, patternmaker's vise.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed neces-

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TABLE III. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 each	II	4.0	Defects per 100 units	4.3.1
Dimensional and weight examination	1 each	S-2	4.0	do.	4.3.2
Testing	1 each	S-2	2.5	do.	4.4.1
	1 each	S-2	4.0	do.	4.4.2
Inspection of preparation for delivery	1 container	S-2	4.0	do.	4.5

sary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table III.

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, coating, construction, workmanship, and marking. Defects are listed in table IV.

TABLE IV. *Classification of defects*

Examine	Defects
Material	Not fabricated of specified material.
Coating	Missing or not as specified, not smooth, uniform or adherent.
Marking	Missing, incorrect, illegible, or not permanent.
Design	Any characteristic not in accordance with specified requirements.
Construction and workmanship	Incomplete, parts missing, not fitted and assembled as required; provisions not made to prevent parts becoming detached while in use.
	Moving parts bind or are excessively loose.
	Threads nonconforming, crossed, stripped, chipped, or torn.
	Not smoothly finished; not free from burrs, fins, or cracks.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 *Tension test.* Each sample vise for which the physical requirements of the steel are specified in 3.6, shall be tested in accordance with method 211 of Fed Std. No. 151.

4.4.2 *Loading test.* Vises shall be subjected to a loading test of 1000 inch pounds. The test shall consist of loading the handle as near the end as practicable while the jaws clamp a parallel block of such dimensions that the jaw opening is equal to the nominal width of the jaws. The test load shall be determined by dividing the required turning moment by a perpendicular distance in inches from the axis of the screw to the line in which the load acts. The handle shall be considered as having taken a permanent set when the maximum deviation of the bent portion from a straightedge held against the longer straight portion is 1/32 inch or more (measurement to be made before and after loading). After the loading test, the vise shall operate smoothly and the component parts shall show no distortion visible to the naked eye.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table V. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of

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closure listed shall be examined on shipping containers fully prepared for delivery

TABLE V. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), the vises shall be preserved, packaged, and packed in accordance with MIL-H-15424. The levels of preservation and packaging shall be A or C and the level of packing A, B, or C as specified (see 6.1).

5.1 1 As part of the requirements of section 5 of this specification, table VI is considered as part of table I of MIL-H-15424.

5.2 Marking.

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein, and include the following data in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of vise required (see 1.2).
(*c*) Size, jaw width, or capacity required (see applicable table).
(*d*) Selection of applicable levels of preservation, packaging, and packing required (see 5.1 and 5.1.1).

6.1.1 When replacement parts are required, the procurement document should specify the following:

(*a*) The name or trademark of the manufacturer of the vise for which the part is required.
(*b*) Proper name or number of the part required.
(*c*) Quantity of each item required.
(*d*) Other data or description applicable to the vise which will assist in identifying the proper replacement parts.

6.2 Supersession data. The classification of vises covered in this specification corresponding to those covered in Federal Specification GGG-V-436d are as follows:

GGG-V-436d	GGG-V-445 a
Type XV—Woodworker's.	Type I—Woodworker's.
Class 1—Continuous screw.	Class 1—Continuous screw.
Class 2—Quick acting.	Class 2—Quick acting.
Type XVI—Patternmaker's.	Type II—Patternmaker's.

TABLE VI

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Vise:					
Woodworking					
up through 20 pounds	P-2, P-5	I	1	8	
20 through 35 pounds	P-2, P-5	I	1	4	
35 through 85 pounds	P-2, P-5	I	1	2	

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MILITARY INTEREST:

CUSTODIANS:	*Review Activities:*	*User Activities:*
Army—GL Navy—Coordination waived Air Force—Coordination waived	GL	EL, MI, MO, WC, CE

Preparing Activity:

GSA-FSS

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